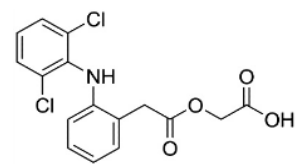


PURPOSE

ACECLOFENAC (ACF)



A glycolic acid ester of diclofenac with four-fold lesser gastric ulcerogenicity

Used in pain and inflammation in arthritic conditions- 100 mg twice daily

Challenges: Poor water solubility, pH dependent solubility, low compressibility, sticking and picking

A sustained release (SR) once a day ACF formulation is desired for better patient compliance

It is critical to choose appropriate excipients for ACF tablets to address above challenges

COMPRITOL® 888 ATO: A Versatile SR Lipid Matrix Forming Agent

Chemical composition

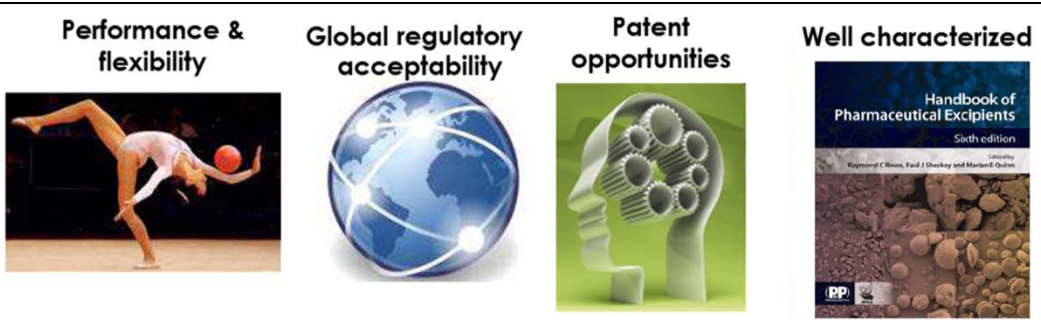
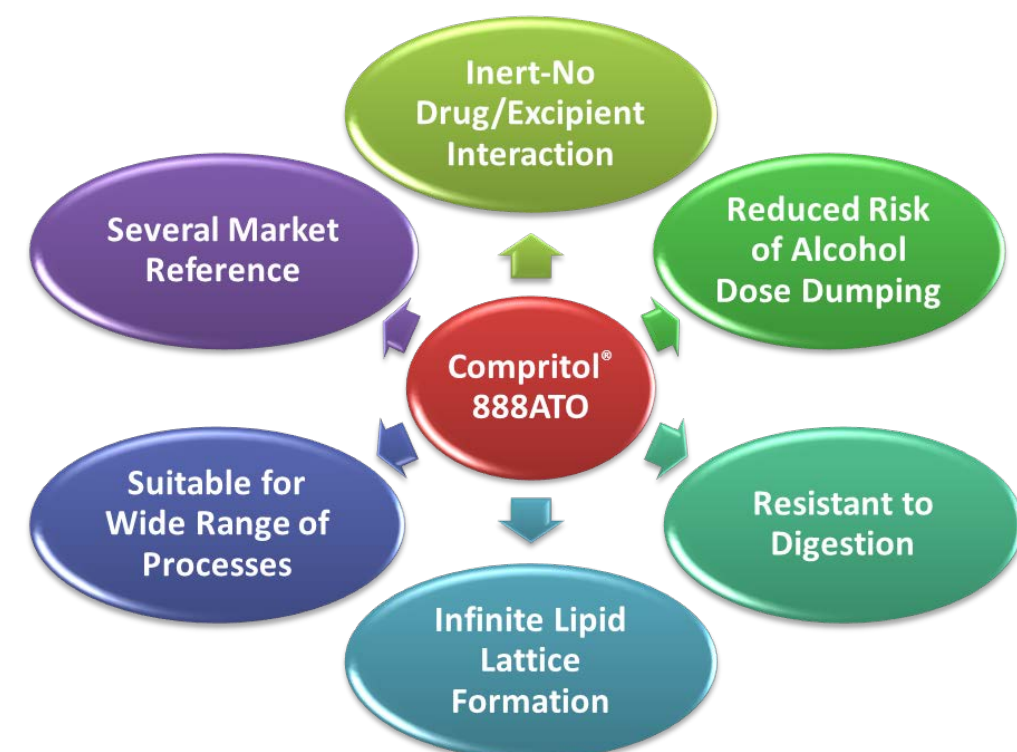
- Mixture of mono-, di- and tribehenate esters of glycerol
- 18%
- 52%
- 28%

Physical properties

- Fine, white atomized powder of 50µm
- Spherical particles
- Drop point 69 – 74°C
- HLB 1, water insoluble
- Stable over 3 years at < 35°C

Regulatory information

- Conforms to USP/NF/EP/Chinese Ph.
- GRAS, FDA IIG, List of Acceptable Non-medicinal Ingredients



METHOD

WET GRANULATION

ACF-SR tablets were prepared using Compritol® 888ATO as lipid matrix. Table-1 and Fig.-1

To address processing issues various ratios and combinations of Compritol® 888ATO, diluents, lubricants and glidants were evaluated and best compositions are shown in Table-1

Sifting of intragranular ingredients through ASTM #40

ACF + MgO and Compritol® 888ATO - Blending

Addition of other intragranular ingredients - Blending

Wet granulation using PVP K30 solution in RMG followed by drying at 40°C

Table 1: Formulations

Ingredients	Batch-1		Batch-2	
	% weight	mg/ tab	% weight	mg/ tab
INTRAGRANULAR				
Aceclofenac IP	50.76	200.00	53.33	200.00
Magnesium oxide	1.27	5.00	1.33	5.00
Compritol® 888 ATO	10.15	40.00	9.33	35.00
Avicel® PH101	22.84	90.00	-	-
Mannitol	5.84	23.00	17.33	65.00
Dicalcium Phosphate	-	-	4.00	15.00
PVP K30	3.05	12.00	2.67	10.00
Purified Water	qs	qs	qs	qs
Intragranular Total	93.91	370.00	88.00	330.00
EXTRAGRANULAR				
Syloid® 244FP	2.03	8.00	2.13	8.00
Avicel® PH200	-	-	5.60	21.00
Talc	2.03	8.00	2.13	8.00
Magnesium stearate	2.03	8.00	2.13	8.00
Total Tablet Weight	100.00	394.00	100.00	375.00

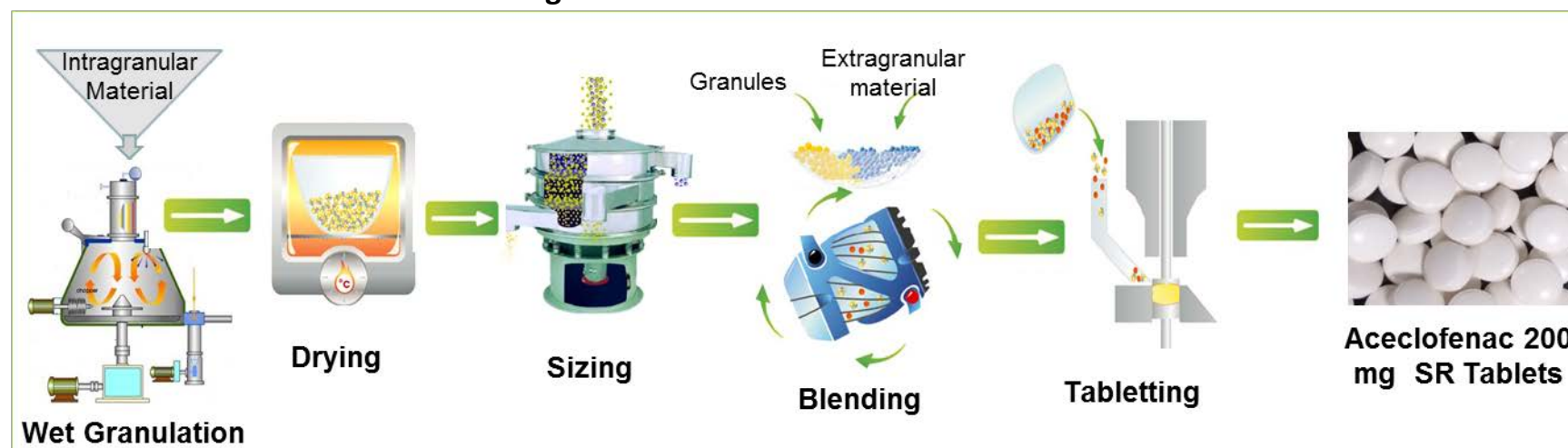
Table 2: Characterization of Granules

Parameters	Batch-1	Batch-2
Moisture content (%)	1.68	1.83
Bulk density (g/mL)	0.425	0.556
Tapped density (g/mL)	0.5	0.654
% Compressibility index	14.97	15.00
Hausner's ratio	1.17	1.18

Compression using 'D' tooling, 9.5 mm round shaped concave, plain punches on fully automated rotary tablet compression machine (Eliza Press)

White colour round shaped tablets were obtained with shiny surface

Fig. 1: Schematic Presentation of Process



EVALUATION OF TABLETS

Physical Evaluation

- Weight variation, Thickness, Hardness, Friability,

Dissolution Study

- Dissolution in 900mL Phosphate buffer pH7.4, USP Type-II

Alcohol Dose Dumping Study

- Dissolution in 900 mL 0.01N HCl with 10% and 40% ethanol

Analysis

- UV spectrophotometry, λ_{max} 273 nm

Curing Effect

- Curing of tablets at 50°C for 24 h-to study its effect on dissolution

RESULTS

Successful wet granulation of ACF with Compritol® 888ATO was accomplished

Magnesium oxide showed improvement in the flow properties of ACF- static charge neutralization

Granules after sifting through ASTM#25 showed good flow properties

Extragranular excipients further improved the flow property of the granules thus alleviating processing issues

Evaluation of ACF tablets for various parameters are depicted in Table-3

Table 3: Evaluation of Tablets

Test	INITIAL		Post Curing at 50°C for 24 h	
	Batch -1	Batch-2	Batch -1	Batch-2
Weight of tablets (mg) (n=20)	394±2.0	375±3.2	394±2.2	375±2.8
Thickness (mm)(n=10)	5.3±0.2	5.0±0.2	5.3±0.2	5.0±0.2
Hardness (N)(n=10)	120±10	90±10	140±10	110±10
%Friability (n=20)	0.038	0.046	0.028	0.034
%Drug content (n=10)	99.8±0.6	99.96±0.6	99.9±0.4	99.96±0.6

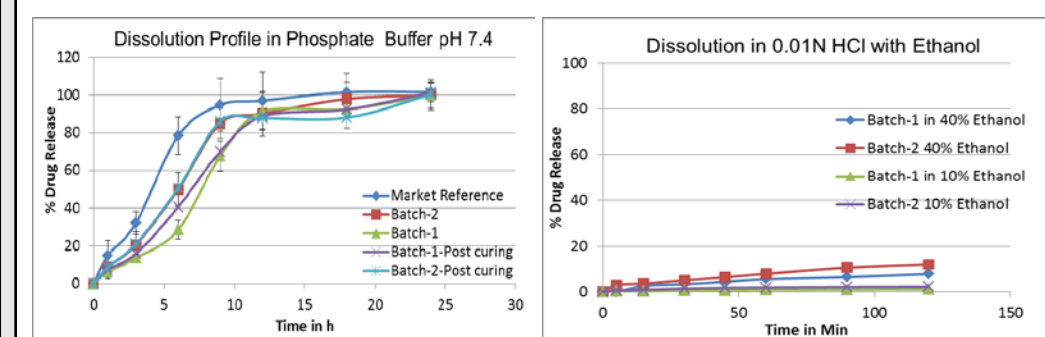
Dissolution of ACF 200 mg tablets of batch-1 and 2 exhibited sustained release up to 18 h (Fig. 2)

Curing of tablets did not affect the dissolution profile, this may be attributed to alkaline microenvironment generated by MgO (Fig. 2)

The release profiles were comparable to a marketed sustained release tablet formulation (Fig. 2)

No alcohol dose dumping was observed in 0.01N HCl with 10% and 40 % ethanol (Fig. 2)

Fig. 2: Dissolution Profile and Dose Dumping Study



CONCLUSION

Compritol® 888ATO as release modifier resulted in successful development of ACF SR 200 mg tablets for once a day pain treatment.

The developed formulation was scaled-up by wet granulation

Alcohol dose dumping was not evident

Curing of tablets was not a necessity

ACKNOWLEDGEMENT

- Grace Davison Chemicals India Pvt. Ltd. for Syloid® 244FP.
- BASF Ltd. for Kollidon® 30.
- Knowell Pharmsolutions for Mannitol.

REFERENCE

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